

Motor selection (Second approach)

Weight of each subsystem:

- Chassis subsystem = 6.1 Kg
- Scissor lift mechanism = 7.5 kg
- Package gripping Mechanism = 7.5 Kg
- Payload = 30 Kg

Total weight = 51.1 Kg

1. Let the nominal speed (V_{max}) = 0.5 m/s
2. Time for acceleration = 3 sec. $\rightarrow a = V_{max}/T_a = 0.5/3 = 0.16\text{m/sec}^2$
3. Wheel's diameter (D_w) = 130mm = 0.13m
4. Thickness of the wheel (T_w) = 59mm = 0.059m
5. Mass of the wheel (M_w) = 173 gm = 0.173 kg
6. Weight of the wheel (W_w) = $0.173 * 9.81 = 1.7 \text{ N}$

Calculate the traction wheel rotation speed :

- $N_t = (60 * V_{max}) / \pi * D_w = 73.45 \approx 74 \text{ RPM}$

Calculate the thrust force (F_t):

- $F_t = U * g(\text{total load}) = 17.54\text{N}$
 - $g \rightarrow$ acceleration Due to gravity = 9.81
 - $U \rightarrow$ coefficient of friction = 0.035 (needs to be reconsidered)

Power Required:

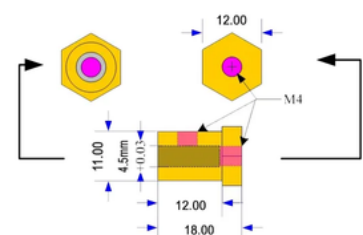
- $P_u = F_t * V_{max} = 8.77 \text{ watt}$, therefore each motor of the four drive motors needs **2.2 Watt** to operate

Specifications of the wheels:

- Outer diameter of wheel: 130mm
- Thickness: 59mm
- Brass coupler inner diameter: 4mm
- Interior six-angle side to side for 12mm (for Hex Coupler)
- Weight: 173 gr (each wheel with mounting hub)

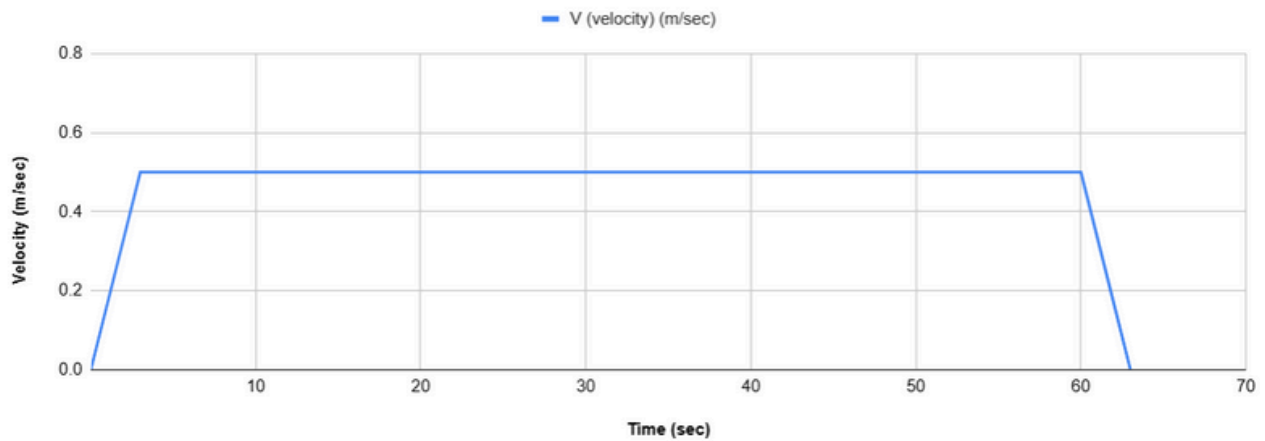


Wheels selected



Wheel Coupler

Motion Profile



- $T_a = T_d = 3 \text{ sec.}$
- $T_c = 60 \text{ sec}$ (average time to deliver a package in a warehouse)
- Angular velocity ($W_{\max}$) = $V_{\max} / \text{Radius of wheel} = 0.5 / 0.065 = 7.7 \text{ rad/sec}$
- Angular acceleration (α) = $W_{\max} / T_a = 7.7 / 3 = 2.6 \text{ rad/sec}^2$

Inertia of the wheel:

- Inertia of the wheel: $J_{\text{wheel}} = [(W_w \cdot r^2)] / 2 \cdot g = 0.0003645 \text{ N.m.sec}^2$
- $J_{\text{load}} = [W_l / g \cdot e] \cdot r^2 = 51.1 / 9.8 \cdot 0.65 \cdot 0.065^2 = 0.033 \text{ N.m.sec}^2$
- $J(\text{system}) = J_{\text{wheel}} = 0.04 \text{ N.m.sec}^2$
 - $e \rightarrow$ efficiency

Torque required for movement:

- Torque due to friction : $T_c = 51.1 / 4 \cdot 9.81 \cdot 0.035 \cdot 0.065 = 0.314 \text{ N.m}$
 - total load = 51.1 Kg
 - $g = 9.81 \text{ m/sec}^2$
 - U (coefficient of friction) = 0.035 (needs to be reconsidered)
 - Radius of the wheel = 0.065 m
 - Divide the load by four as the load is equally distributed on 4 drive motors
- Torque due to acceleration : $T_a = \alpha \cdot J(\text{system}) = 0.11 \text{ N.m}$

$$T_{\text{RMS}} = \sqrt{\frac{(T_a + T_c)^2 \times t_a + T_c^2 \times t_c + (T_d + T_c)^2 \times t_d + T_h^2 \times t_h}{t_a + t_c + t_d + t_h}}$$

$T_{\text{rms}} = 0.32 \text{ Nm} = 3.3 \text{ Kg.cm}$

